

A terminal on Γ in Jochum's formulation — derivation report

Status: derivation in progress. §§1–5 are worked and I believe them. §6 is a proposed resolution with a dimension count that balances but is **not yet verified**, and §7 says what would verify it. Nothing here is implemented.

Why this document exists: `tools/gauge_spike/RESULTS_SPIKEC.md` milestone 2 found that Jochum, Farle & Dyczij-Edlinger's formulation — the only one in the literature carrying a materially-weighted gauge interface condition at a conductor/insulator boundary, which is exactly where our failing port sits — has **nowhere to put a port**. This works out why, and what would have to change.

1. The formulation, in our convention

Jochum writes $e^{+ik_o \cdot}$ with a scaled potential; translated to ours ($e^{+j\omega t}$, $E = -j\omega A - \nabla\Phi$), with $\kappa = \sigma + j\omega\epsilon$ and $v = 1/\mu$:

$$\begin{aligned} A &= A_c + \nabla\psi, & B &= \nabla \times A_c, & (15), (16) \\ E &= -\nabla V - j\omega(A_c + \nabla\psi) & & & (17) \end{aligned}$$

Three unknowns: A_c on cotree edges, ψ nodal, V nodal. A_c lives in the reduced space of the **inexact Helmholtz splitting** (14), realized discretely by tree–cotree — so the tree is still there, but it decomposes the space rather than fixing the gauge.

In the lossy sub-domain Ω_C :

$$\begin{aligned} \nabla \times (v \nabla \times A_c) + \kappa [j\omega(A_c + \nabla\psi) + \nabla V] &= J_i & \text{Ampère} & (18a) \\ \nabla \cdot [\kappa(A_c + \nabla\psi)] &= 0 & \text{GAUGE} & (18b) \end{aligned}$$

In the lossless sub-domain Ω_N :

$$\begin{aligned} \nabla \times (v \nabla \times A_c) + j\omega\epsilon [j\omega(A_c + \nabla\psi) + \nabla V] &= J_i & \text{Ampère} & (19a) \\ \nabla \cdot \{ \epsilon [j\omega(A_c + \nabla\psi) + \nabla V] \} &= \rho & \text{Gauss} & (19b) \end{aligned}$$

Since $\kappa|_{\Omega_N} = j\omega\epsilon$, (19a) is (18a) with the local κ — one expression covers both. The asymmetry is entirely in the **second** equation: a gauge in the conductor, physics in the insulator.

2. Which equation lives in which row — the fact everything turns on

Jochum tests each equation against a different space (his §5.2):

WEAK FORM	EQUATION TESTED	TEST FUNCTION	PRODUCES THE ROW FOR
(21)	Ampère (18a)	$w_c \in \tilde{Q}H(\text{curl})$	** A_c **
(22)	Ampère (18a)	** $\nabla\psi$ **, $\psi^- \in H^1_o$	** ψ **
(23)	gauge (18b)	$\bar{V} \in H^1_o$	** V **

and states that (22) is a weak form of the **continuity equation**.

So, and this is the crux:

** ψ 's row is continuity. V 's row is the gauge.**

That cross-placement is what makes the off-diagonal blocks transposes and the matrix complex-symmetric — it is a feature, not an accident. But it **inverts the usual A - Φ correspondence**, where a current-driven port's row is the continuity equation and its unknown is the potential.

3. Deriving (32) without a terminal

Test (18b) by $\bar{V}$ over Ω_C and integrate by parts:

$$-\int_{\Omega_C} \kappa(A_c + \nabla\psi) \cdot \nabla\bar{V} + \oint_{\partial\Omega_C} \bar{V} \hat{n} \cdot [\kappa(A_c + \nabla\psi)] = 0$$

Test (19b) by $\bar{V}$ over Ω_N , noting $\epsilon[j\omega(A_c + \nabla\psi) + \nabla V] = -\epsilon E$:

$$-\int_{\Omega_N} (-\epsilon E) \cdot \nabla \bar{V} + \oint_{\partial\Omega_N} \bar{V} \hat{n} \cdot (-\epsilon E) = \int \rho \bar{V}$$

Let $\hat{n}$ point from Ω_C into Ω_N , so $\hat{n}_C = \hat{n}$ and $\hat{n}_N = -\hat{n}$ on Γ . The two surface integrals contribute, on Γ ,

$$\oint_{\Gamma} \bar{V} \{ \hat{n} \cdot [\kappa(A_C + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N \} dS \quad (\star)$$

With $\bar{V}$ single-valued and **free** on Γ , $\bar{V}$ is arbitrary there, so $(\star)$ vanishes only if its integrand does pointwise:

$$\hat{n} \cdot [\kappa(A_C + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N = 0 \quad \text{on } \Gamma \quad (32)$$

This is Jochum's (32), and it is not an extra assumption — it is the condition under which the two sub-domain weak forms assemble into one system with no leftover surface term. It is materially weighted on both sides: $\kappa = \sigma + j\omega\epsilon$ from the conductor, ϵ from the insulator. That weighting is why §12.3 ranks it above the ϵ -blind Coulomb-family conditions.

4. What a terminal is, formally

A current-driven terminal on $\Gamma_T \subseteq \Gamma$ is two statements:

1. **$\bar{V}$ is single-valued on Γ_T** — one unknown $\bar{V}_T$, not one per node. In the discrete setting this is the aggregation $A = P \hat{A} P^T$ (Stysch §6.6.2, [GAUGE_CHOICE.md](#) §11.10.5), which is a congruence and so preserves symmetry.
2. **The total current through Γ_T is prescribed**:

$$I = \int_{\Gamma_T} \hat{n} \cdot (\kappa E)_C dS \quad (33)$$

Statement 1 collapses the test space: on Γ_T , $\bar{V}$ is a single constant rather than an arbitrary function.

5. The obstacle, located exactly

With $\bar{V}$ constant on Γ_T , $(\star)$ no longer forces its integrand to vanish there. It forces only the **integral**:

$$\int_{\Gamma_T} \{ \hat{n} \cdot [\kappa(A_c + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N \} dS = 0 \quad (34)$$

That is one scalar equation, which is right — one equation for one unknown V_T . But now look at what it contains against what we need to impose:

NORMAL FLUX APPEARING

(34), the V_T row $\hat{n} \cdot (\kappa \star A \star)_C$ — the **gauge** flux

(33), the current $\hat{n} \cdot (\kappa \star E \star)_C$ — the **current**

These are different physical objects. A is not E . The gauge equation's natural boundary datum is the normal component of κA , which is a statement about the gauge, and no choice of right-hand side turns it into a current.

And the current flux $\hat{n} \cdot (\kappa E)$ does appear in the weak form — but in the $\star \psi$ row, since (22) is continuity and its surface term is $\oint \hat{n} \cdot (\kappa E) \psi$.

So the terminal's two requirements land in two different rows:

the terminal POTENTIAL we want to read	is	V_T	
the row V_T owns	is	the GAUGE	(34)
the row carrying the CURRENT	is	ψ 's	(continuity)

That is the whole obstacle, and it is structural rather than a matter of assembly detail.

5.1 It explains the measurement exactly

Spike C milestone 2 put the current drive on V 's row — the natural thing to do, and wrong here. Measured consequence: a terminal potential of $-1.149425e-05$ V identical across three trees to $2.4e-14$, with **zero reactance, unchanged from 50 Hz to 50 GHz**, and a V-coupling of $G_{\text{eps}N} = 4.2e-02$ against gauge terms $G_{\text{kap}} = 7.5e+07$ — a factor $5.7e-10$.

Every one of those follows from §5: the drive perturbed the gauge condition instead of injecting current, leaving a decoupled DC conduction problem whose solution is linear in z and therefore exact in P1. The spike did not fail; it measured this.

5.2 Why his paper never hit it

His §6.1 cavity sets $\phi = 0$ and $\phi = 1$ on faces, and §6.2 is a voltage-driven RLC field model. Both are **Dirichlet conditions on V **, which are trivial here — prescribe V and the row disappears. **A voltage port in Jochum's formulation is free; only a current-driven floating terminal is a problem.** That is a useful narrowing: the gap is one port type, not the whole concept.

6. Proposed resolution — aggregate ψ as well as V

Not verified. The dimension count balances and the physics is plausible; that is all that can be claimed so far.

If Γ_T needs two conditions — the gauge (34) and the current (33) — then give it **two unknowns**: aggregate ψ on Γ_T as well as V .

$V_T = V$	on Γ_T	one unknown	row: aggregated gauge	(34)
$\psi_T = \psi$	on Γ_T	one unknown	row: aggregated continuity = I	(33)

Count. With N_T nodes on Γ_T , before: N_T V -unknowns and N_T ψ -unknowns against N_T gauge rows and N_T continuity rows. After: 2 unknowns against 2 rows. Balanced.

****Legitimacy of aggregating ψ .** ψ is the gauge function. Forcing it constant on a surface is a *gauge choice*, not a physical constraint — it is precisely what contracting a surface in the tree does (§15.9), and what $\hat{n} \times A = 0$ achieves on the outer boundary. So it costs nothing physical.

Symmetry. Both aggregations are congruences $P(\cdot)P^T$, applied to the V and ψ blocks. A congruence preserves complex symmetry, so the system should remain symmetric — the property §13.2 identifies as Jochum's main advantage over Chew.

****What V_T then means.** V_T is the terminal potential, read as the answer. ψ_T is a gauge constant on the terminal, carrying no physical meaning — the analogue of $\psi = 0$ on the outer boundary.

6.1 What is not yet established

- **Uniqueness.** Does (33)+(34) determine (V_T, ψ_T) uniquely, or is the pair

singular in some limit? The count balances; that is necessary, not sufficient.

- **Whether (32) survives off Γ_T .** The derivation in §3 assumed $\bar{V}$ free on

all of Γ . On $\Gamma \setminus \Gamma_T$ it still is, so (32) should hold there — but the two regions share an edge, and the rim of Γ_T needs checking.

- **The low-frequency limit.** $\kappa \rightarrow \sigma$ in the conductor and $\kappa \rightarrow 0$ in the

insulator as $\omega \rightarrow 0$. §11.10 measured that an insulator-interior ψ row vanishes at DC, which is why Stysch's §5.5 restricts ψ to the interface. Γ_T is on the interface, so ψ_T should survive — but that is an expectation, not a measurement.

- **Whether I belongs on ψ_T 's row with a factor.** (33) is the current;

(22) is continuity tested by $\nabla\psi$. The aggregated continuity row equals the net current through Γ_T only up to a sign and possibly a scaling from the non-dimensionalization. That needs doing carefully, and `RESULTS_COMPARE.md` records what dropping such a factor costs.

7. How to test it, cheaply

`tools/gauge_spike/spikeC.py` already has everything but this: a validated miniature (milestone 1: R exact to 0.06 %, terminal ϕ moving 3.8 % with the tree), a Jochum assembly (milestone 2), and a terminal aggregation.

The change is to `solve_jochum`: add a second aggregated unknown on the terminal face for ψ , move the drive from V 's row to ψ 's, and re-run. The acceptance test is unchanged and already written:

	MUST BE
R	11.494253 $\mu\Omega$ — the analytic DC value
reactance	present and frequency-dependent, not constant across decades
terminal ϕ across three trees	invariant
symmetry residual	$\sim 1e-16$

Milestone 2's failure signature is specific enough to be diagnostic: if the reactance is again zero and frequency-independent, the drive is still in the gauge row and §6 is wrong.

8. If §6 fails

Two fallbacks, both more expensive:

- **Re-derive (32) with the terminal present from the start**, rather than

aggregating afterwards. (32) came from requiring the surface terms to cancel; with a terminal, the correct statement may be a *modified* interface condition on Γ_T rather than the same one aggregated. This is the honest version of "the derivation", and §6 is the shortcut worth trying first.

- **Take the condition from outside.** A boundary-integral exterior supplies the

relation between A and Φ on a coupling surface from the exterior solution, which is what `GAUGE_CHOICE.md` §0 wants Chew for. That removes the need to invent an interface condition at all, at the cost of the whole BI programme.

9. Sources

Jochum, Farle & Dyczij-Edlinger (2016) , <i>A Symmetric and Low-Frequency Stable Potential Formulation...</i> , in <i>Scientific Computing in Electrical Engineering</i> (SCEE 2014), Mathematics in Industry 23 , 63–71, doi:10.1007/978-3-319-30399-4_7 — <code>APhi_Papers/Jochum_SymmetricLowFreqStablePotentialFormulation.pdf</code>	Equations (14)–(19), (21)–(23), (32); §5.2's test-function assignment, which §2 above turns on; §6's examples, which are voltage-driven and so never meet this problem.
Stysch (2022) , PhD thesis, TU Darmstadt	§6.6.2's aggregation <code>A = P</code> <code>A P^T</code> , which §4 and §6 use for the terminal; §5.5's restriction of <code>ψ</code> to the interface, which §6.1 relies on.
<code>tools/gauge_spike/RESULTS_SPIKEC.md</code>	Milestone 2's measurement, which §5.1 explains.
<code>GAUGE_CHOICE.md</code> §13	The reading of Jochum this builds on.